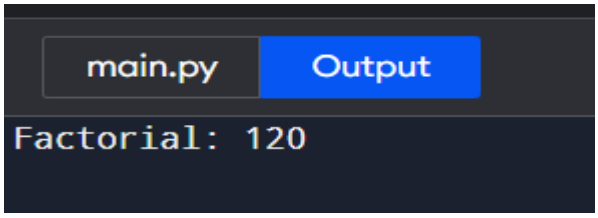
#Recursion

1. Write a recursive function to calculate the factorial of a number.

```
def factorial(n):
    if n == 0 or n == 1:
        return 1
    else:
        return n * factorial(n - 1)

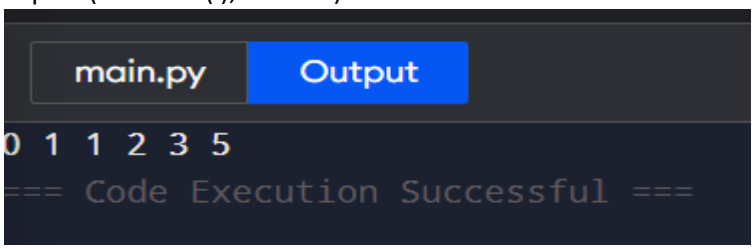
print("Factorial:", factorial(5))
```



2. Write a recursive function to generate the Fibonacci series up to n terms.

```
def fibonacci(n):
    if n <= 1:
        return n
    else:
        return fibonacci(n - 1) + fibonacci(n - 2)

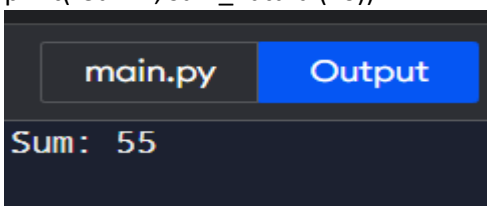
n = 6
for i in range(n):
    print(fibonacci(i), end=" ")
```



3. Create a recursive function to find the sum of natural numbers up to n.

```
def sum_natural(n):
    if n == 0:
        return 0
    else:
        return n + sum_natural(n - 1)

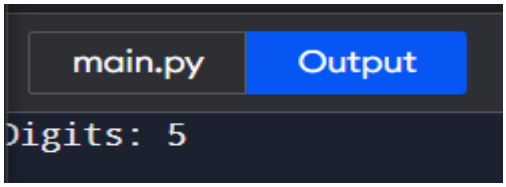
print("Sum:", sum_natural(10))
```



4. Write a recursive function to reverse a string.

```
def count_digits(n):
    if n == 0:
        return 0
    else:
        return 1 + count_digits(n // 10)

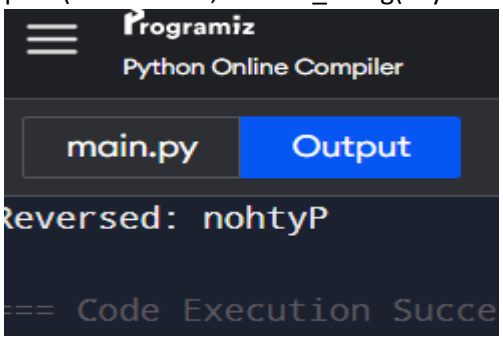
print("Digits:", count_digits(12345))
```



5. Write a recursive function to count the digits in a number.

```
def reverse_string(s):
    if len(s) == 0:
        return ""
    else:
        return s[-1] + reverse_string(s[:-1])

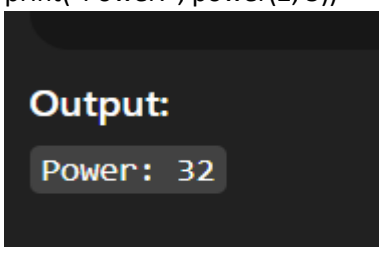
print("Reversed:", reverse_string("Python"))
```



6. Create a recursive function to calculate the power of a number (x^y).

```
def power(x, y):
    if y == 0:
        return 1
    else:
        return x * power(x, y - 1)

print("Power:", power(2, 5))
```

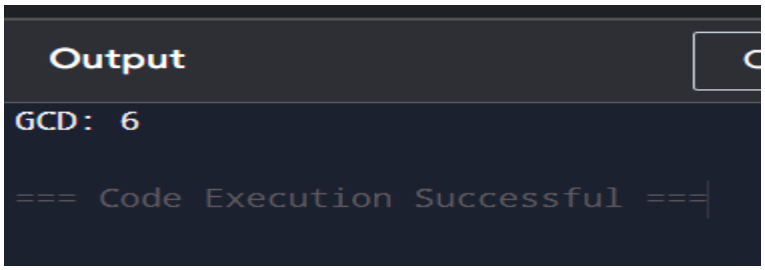


7. Write a recursive function to find the greatest common divisor (GCD) of two numbers.

```
def gcd(a, b):
    if b == 0:
        return a
    else:
```

```
    return gcd(b, a % b)

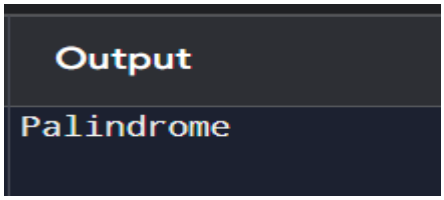
print("GCD:", gcd(48, 18))
```



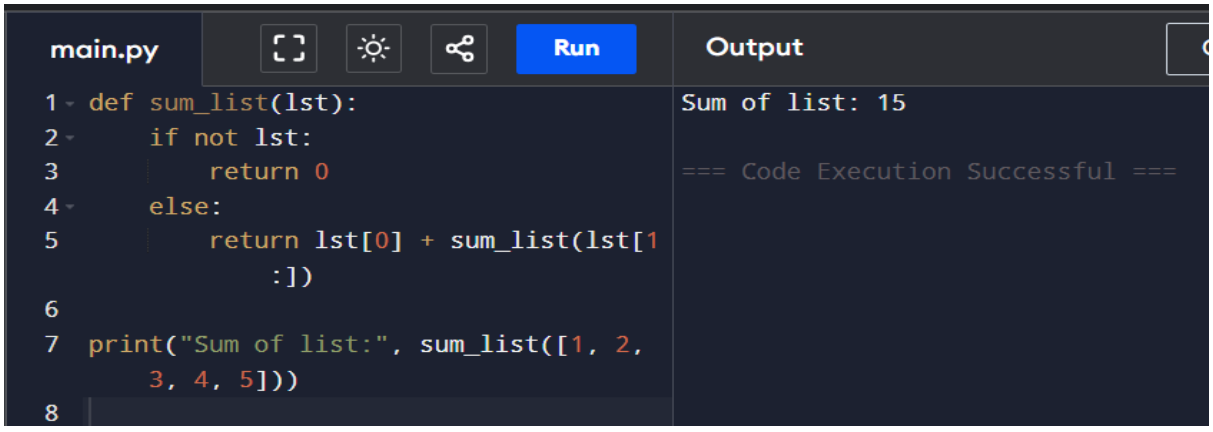
8. Create a recursive function to check whether a number is palindrome or not.

```
def is_palindrome(num, temp=None):
    if temp is None:
        temp = num
    if num == 0:
        return 0
    rev = (num % 10) * (10 ** (len(str(temp)) - len(str(num)))) + is_palindrome(num // 10, temp)
    return rev

num = 121
if num == is_palindrome(num):
    print("Palindrome")
else:
    print("Not Palindrome")
```



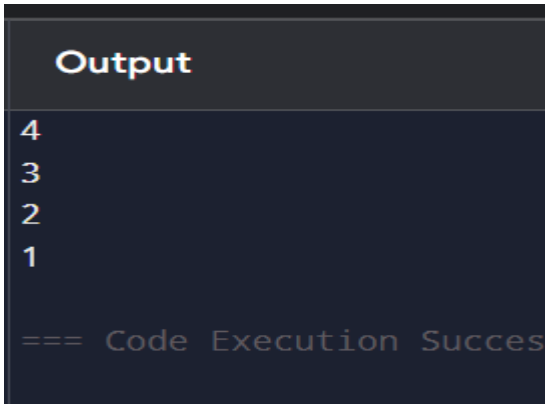
9. Write a recursive function to find the sum of elements in a list.



10. Write a recursive function to print all elements of a list in reverse order.

```
def print_reverse(lst):
    if len(lst) == 0:
        return
    print(lst[-1])
    print_reverse(lst[:-1])
```

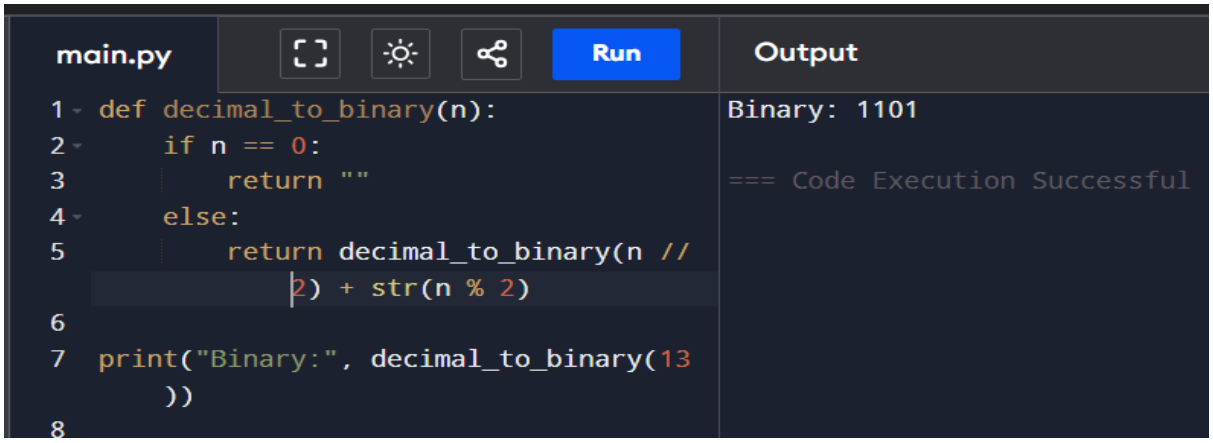
print_reverse([1, 2, 3, 4])



11. Create a recursive function to find the length of a string.



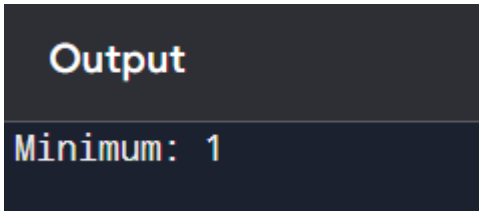
12. Write a recursive function to convert a decimal number to binary.



13. Create a recursive function to find the minimum element in a list.

```
def find_min(lst):
    if len(lst) == 1:
        return lst[0]
    else:
        m = find_min(lst[1:])
        return m if m < lst[0] else lst[0]

print("Minimum:", find_min([5, 2, 8, 1, 4]))
```



14. Write a recursive function to compute the sum of digits of a number.



15. Create a recursive function to check whether a given string is palindrome.

```
def is_string_palindrome(s):
    if len(s) <= 1:
        return True
    elif s[0] != s[-1]:
        return False
    else:
        return is_string_palindrome(s[1:-1])

print("Palindrome:", is_string_palindrome("madam"))
```

